

Matching Algorithm: Summary so far

$G = (V, E)$,

M

X exposed vertices

M -alternating trees ; Maximal M -alt. trees

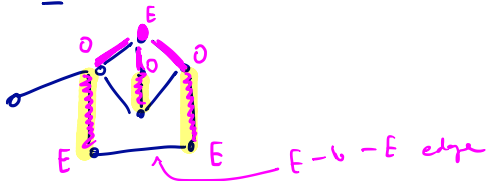
Lemma: If the alg. in prev. lecture for maximal M -alt. trees

and:

1. It did not find an M -aug path

2. There are no edges b/w " E ",

Then M is a maximum size matching.



Proof of Sufficient Condition

set of nodes labeled "E"

A. Any edge in G with an endpoint in E must have other endpoint in O .

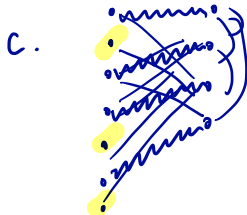
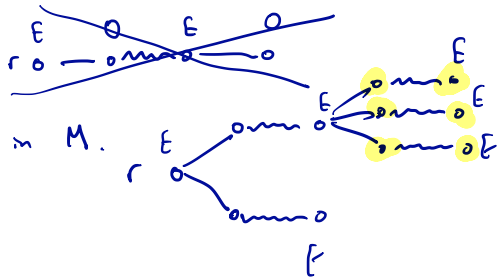
$E-E$ ruled out by assumption.

$E-O$ ruled out by maximality

$$B. |Even| = |Odd| + |X|$$

$\Rightarrow$ all paths end in E , and all E nodes except r are matched to O nodes by edge in M .

b/c no M -aug path

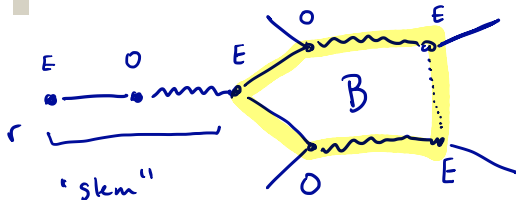


Bipartite Graphs

Thm: If $\text{alg}^{\text{for } M\text{-maximal}}$ alt trees finds
a collection of trees w/o
terminating at an M -ary path,
then M is max size.

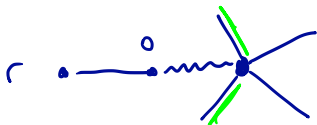
Exercise: Work out details.

Blossoms — Even to Even Edges



odd cycle b/c we have E-E
"Blossom" edge.

Contracting blossom: G/B



Key: We can now grow the
free in new ways!